

Poster Presentation Speaker Notes

Evaluating LLMs’ Contextual Understanding with Knowledge Graphs

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Opening

*“Good morning / afternoon. Our project is titled **Evaluating LLMs’ Contextual Understanding with Knowledge Graphs**. The core question we are addressing is — LLMs answer questions well, but do they truly understand the context given to them, or are they just producing plausible-sounding text?”*

Problem Statement

*“When we look at existing work on LLM contextual understanding, they focus on things like conversation tracking — whether the model remembers what was said earlier — or coreference resolution — finding who refers to whom. But what we care about is specifically in **Question Answering**: given a question and a supporting context, does the model actually utilise that context to form its answer?”*

“The existing metrics like BLEU and ROUGE only do n-gram level word checking — surface level. They cannot tell whether the model understood the context or just happened to repeat some words. We need something semantic.”

Research Objectives

“So we set three goals. First, build a KG-based methodology — because representing text as subject-predicate-object triples means varying word choices will not fool the metric. Second, define similarity metrics that check both factual accuracy and how well the model utilised the context. Third, go deeper — detect exactly what kind of fact deviations occur when the model misses or misrepresents something.”

Methodology — Overview

“Our pipeline works like this. We take the question with its context, collect the LLM’s answer, and also have the ground truth answer. We construct three knowledge graphs — one from the LLM answer, one from the ground truth, one from the context. We then compare these using S3KG. For the hardest cases — the lowest-scoring 5% — we run the Triplet Analyzing Unit to diagnose specific errors.”

Methodology — KGC (Knowledge Graph Construction)

“Building consistent KGs is not trivial. Standard extraction produces inconsistent labels across different texts — “Einstein” in one graph, “A. Einstein” in another — making comparison meaningless.”

“We use a method inspired by KRPO — Knowledge Restoration-driven Prompt Optimization.

*Instead of extracting each KG separately, we extract all three in a **single joint prompt** so they share the same label space from the start.”*

*“Each extracted triplet is then scored using an **NLI model** — Natural Language Inference — which determines the logical relationship between two sentences. It classifies each triplet as Entailment, Neutral, or Contradiction against its source text. Entailment gives +1, Neutral 0, Contradiction −0.5.”*

“We compute a combined score across all three graphs. If it is above the threshold we accept. Otherwise, we do joint backward evaluation — identify where the prompt is causing alignment issues — generate feedback, rewrite the prompt, and retry. The new prompt is only accepted if it scores better than the previous one.”

Methodology — Triplet Analyzing Unit

“The Triplet Analyzing Unit operates on the lowest-scoring pairs. We take the Gold KG and LLM KG, convert each triple to a sentence, and embed it with a SentenceTransformer. We compute cosine similarity — if it is above 0.65, the triples are considered aligned.”

“For the residual — unaligned triples — we do component-wise comparison of Head, Relation, and Tail to classify the error type:”

- Head and Tail match but Relation differs → **Relation wrong**
- Relation matches but entities differ → **Entity wrong**
- Exists only in the LLM KG → **Extra triplet** (hallucination)
- Exists only in the Gold KG → **Missing triplet** (omission)

S3KG — Semantic Structural Similarity

“Now for our core metric — S3KG: Semantic Structural Similarity.”

*“Given KG_1 and KG_2 , we first do **triple-level matching** — embed every triple in KG_1 using SBERT, find the best-matching triple in KG_2 by cosine similarity, and filter KG_2 down to the same size as KG_1 . This is important — it anchors the comparison to what the LLM actually said.”*

*“Next, **soft label alignment** — for every node and edge label, if its SBERT similarity to a label in the other graph is above 0.65, we relabel it to a common anchor. We do entities and relations separately to avoid cross-type collisions.”*

*“Then we run the **WL Kernel** on the relabelled graphs — this aggregates neighbourhood labels across 5 iterations to get a structural similarity score S_{WL} .”*

*“Finally, **SBERT mean-pool** — we embed all triples in each KG and take the mean, then compute cosine similarity for a semantic score S_{SBERT} .”*

“The final score: $S_{S3KG} = \alpha \times S_{WL} + (1 - \alpha) \times S_{SBERT}$, with $\alpha = 0.5$ giving equal weight to structure and semantics. We selected this α value from our benchmarking.”

LLM Contextual Understanding Evaluation

“We apply *S3KG* across three KG pairs per QA sample. **GoldSim** measures factual accuracy — LLM KG versus Gold KG. **CtxSim** measures contextual faithfulness — LLM KG versus Context KG. **CUS** is their harmonic mean, so a model cannot hide weakness in one dimension by excelling in the other.”

“Looking at results — *Mistral-7B* leads *CUS* on both datasets. *CtxSim* is higher than *GoldSim* across all 8 model-dataset pairs, meaning models elaborate on context rather than produce concise reference-style answers. There is roughly a 10 percentage point gap between *MesaQA* and *PubMedQA* — that is the domain complexity of biomedical vocabulary, not a model size issue. *Falcon-7B* is the weakest across all metrics.”

Benchmarking

“We benchmarked *S3KG* against 7 baselines — lexical methods like *ROUGE* and *BLEU* which do *n*-gram checking, and semantic methods like *BERTScore*, *MiniLM*, and *sentence-T5*. We used 9 datasets — short text pairs, KG-perturbed paragraphs where entity relationships are deliberately swapped, and a Wikipedia entity-swap control.”

“*S3KG* leads on 7 out of 9 datasets in both *F1* and *AUROC*. The biggest gains are on KG-perturbed sets — these datasets are specifically built to catch KG-level errors that traditional metrics are blind to. *ROUGE-1* completely collapses on adversarial datasets — near-random *AUROC*. *Sentence-T5* is the closest rival but fails when relations are perturbed.”

“For *KGC*, our joint extraction approach gives the best balance of accuracy and label alignment — harmonic mean 0.591 versus 0.530 for *KRPO* and 0.576 for the *KEA* baseline.”

“For the Triplet Analyzing Unit, we benchmarked against 3 medical-student annotators — 94% inter-annotator agreement. Our method achieves Micro *F1* of 0.895 versus *KEA*’s 0.836.”

Conclusion

“To summarise — we present **CUS** as a reproducible, interpretable metric for LLM contextual understanding. Our **KGC** method provides the best balance of accuracy and label consistency for downstream multi-graph comparison tasks. **S3KG** achieves top-ranked performance, outperforming existing methods on context-rich datasets. And **Mistral-7B** demonstrates the strongest contextual understanding among same-parameter models.”

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